

Operationalizing the Market for Housing Regulation

An Estimation Blueprint: Random-Coefficients Demand, a Discretionary-Regulation Game, and a Spatial-Equilibrium Wrapper

Daniel Posthumus

June 2026 — working note for meeting with A. Guren

Abstract

This note operationalizes the toy model (`toy_model_v5`). It does the one thing that document deliberately leaves open: it specifies *how each primitive becomes an estimating object*, what *econometric machinery* recovers it, and what *has to be true* for the counterfactuals (the Oates wedge; preemption leakage) to be identified. The architecture has three layers: (i) a **demand layer** — a BLP / residential-sorting random-coefficients model that prices the submarket externality and delivers the price-on-quantity object the model needs; (ii) a **regulation layer** — the homevoter’s two-margin choice estimated as a *static game of incomplete information* among co-submarket jurisdictions, whose strategic-interaction slope is the empirical signature of the spillover; and (iii) a **spatial-equilibrium wrapper** that closes prices, wages, and population so the model can be inverted and counterfactuals computed. The note is explicit about where empirical-IO methods (layers i, ii) and quantitative-spatial methods (layer iii) meet, and it flags — in boxed asides — the modeling decisions to settle with Guren before any estimation begins. Notation follows the toy model: jurisdictions j , submarkets $s(j)$, by-right envelope $\bar{z}_j$, discretionary tax τ_j , effective regulation $r_j = g(\bar{z}_j, \tau_j)$.

Contents

1	What “operationalize” means here	2
2	Layer I — Housing demand as a random-coefficients sorting model	2
2.1	Why this exact object, and what the model needs from it	2
2.2	Specification	2
2.3	Identification and the instrument you want to innovate on	3
3	Layer II — The regulation game on two margins	4
3.1	The within-jurisdiction two-stage game (main spec)	4
3.2	The cross-jurisdiction spillover: two readings to choose between	4
3.3	What the spatial-lag reduced form still does (under either reading)	5
3.4	Action type: continuous vs. ordered vs. binary (open)	5
3.5	The preemption exclusion restriction (not yet committed)	6
3.6	Multiplicity and equilibrium selection (open)	6
4	Layer III — Closing the spatial equilibrium	7
4.1	The minimal closure	7
4.2	Inversion and counterfactuals	7

5	How the three layers compose	8
6	The one slide that should drive the meeting	8

1 What “operationalize” means here

The toy model states three empirical objects: the fragmentation distortion (Question 1, the Oates wedge), the preemption leakage (Question 2), and the cross-jurisdiction reaction function (the signature of the spillover). None is a reduced-form regression coefficient; all three are *counterfactual* objects that require recovering structural primitives — household preferences, the homevoter objective, the disamenity function, agglomeration — from equilibrium behavior. So the blueprint is a structural-estimation pipeline, and the design rule from your prior is right: **keep the layers the project is not *about* as standard as possible, and concentrate originality in the regulation layer and its instrument.**

Division of labor across literatures.

- **Demand** is empirical IO: random-coefficients discrete choice (Berry et al., 1995, Berry, 1994) in its residential-sorting form (Bayer et al., 2007, 2004). This is the part you want to flex IO muscles on; it is also the part the literature has most thoroughly standardized (Nevo, 2000, Conlon and Gortmaker, 2020).
- **Regulation** is also empirical IO, but the *games* branch: jurisdictions choosing $(\bar{z}_j, \tau_j)$ best-responding to co-submarket neighbors is a static discrete/continuous game (Bajari et al., 2010a,b, Seim, 2006). This is where the conceptual novelty of the project lives.
- **Equilibrium closure** is quantitative spatial economics: wages, prices, and population are jointly determined across locations and inverted from observed data (Ahlfeldt et al., 2015, Redding and Rossi-Hansberg, 2017). This is the unfamiliar machinery you need to borrow, and the place to borrow rather than innovate.

2 Layer I — Housing demand as a random-coefficients sorting model

2.1 Why this exact object, and what the model needs from it

The toy model uses demand in three places: the submarket price index $p_s = \alpha N_s / H_s$, the agglomeration wage $w_m = w(N)$, and the household disamenity $-\gamma g(D_j)$. For the counterfactuals you need (a) how submarket price responds to quantity supplied — the slope that makes free-riding operative — and (b) heterogeneous willingness-to-pay to locate in a given jurisdiction, because the homevoter’s asset-value motive is exactly the capitalization of those WTPs. A random-coefficients model delivers both: realistic own- and cross-price elasticities (curing the IIA problem that would make every counterfactual mechanical) and a distribution of marginal WTP for jurisdiction attributes.

2.2 Specification

Index households i , jurisdictions (products) j , markets (submarket $\times$ year) t . Indirect utility of i from locating in j :

$$u_{ijt} = \delta_{jt} + \sum_k \sigma_k \nu_{ik} x_{jt}^k + \pi (\text{demographics}_i \times x_{jt}) - \alpha_i p_{jt} + \epsilon_{ijt}, \quad \delta_{jt} = x_{jt} \beta - \bar{\alpha} p_{jt} + \xi_{jt}, \quad (1)$$

where x_{jt} are jurisdiction attributes (structural density, school quality, transit access, job access JA_{jt} , crime, “character” proxies), p_{jt} is the price index, ξ_{jt} is the unobserved jurisdiction quality, $\nu_{ik} \sim N(0,1)$ are random tastes, and $\alpha_i = \bar{\alpha} + \Pi \text{income}_i$ makes the price coefficient income-dependent (the mechanism by which high- θ residents are the natural demanders of low density, matching the toy model’s θ -sorting). Market shares follow the standard mixed-logit integral; the contraction of Berry [1994] inverts observed shares to δ_{jt} .

2.3 Identification and the instrument you want to innovate on

The endogeneity is $\text{Cov}(p_{jt}, \xi_{jt}) \neq 0$: unobserved desirability raises both price and demand. Three instrument families are available, and your stated goal is to contribute a new one:

1. **BLP/differentiation instruments** — characteristics of competing jurisdictions (Berry et al., 1995, Gandhi and Houde, 2019). Cheap, always available, weak on their own here.
2. **Bayer-style boundary discontinuities** — attributes just across a jurisdiction boundary as instruments / controls for ξ (Bayer et al., 2007). Natural in a fragmented-jurisdiction setting and worth using regardless.
3. **A cost-shifter for the price index — your candidate innovation.** A supply-cost shocker that moves p_{jt} but is excluded from ξ_{jt} . Candidates from your own data environment: (a) *construction-cost exposure* interacted with parcel-level developable-terrain shares (a Saiz-style geographic instrument, Saiz, 2010, but at jurisdiction granularity); (b) *the preemption ramp itself* — builder’s-remedy exposure shifts the supply environment for reasons orthogonal to contemporaneous demand (this is the same variation Layer II leans on, so be careful about reusing it — see Remark 1).

Remark 1 (Do not spend your one good instrument twice). *The preemption ramp is the cleanest exogenous shifter you have, and both the demand price endogeneity and the regulation game want it. If it instruments p_{jt} in demand and identifies the reaction-function slope in Layer II, you are asking one source of variation to do two identifying jobs and the exclusion restrictions may collide. Cleanest resolution: use a construction-cost / geography instrument for demand prices, and reserve the preemption ramp for the regulation game. This is a first-order thing to settle with Guren.*

For Guren — Layer I

- **Is full BLP worth it, or is a nested logit / IV demable elasticity the “minimum viable demand”?** The counterfactuals may only need a credible $\partial p_s / \partial H_s$. If so, RC buys realism in substitution but at real cost. (Your call is to do full RC for the IO training value; worth a sanity check that it is not overkill for what the equilibrium needs.)
- **Unit of demand: jurisdiction, or neighborhood-within-jurisdiction?** The toy model’s externality lives at the *submarket* level, but sorting is at a finer grain. Defining products as jurisdictions may be too coarse to estimate substitution; defining them as tracts may detach demand from the regulating unit. The aggregation choice is load-bearing.
- **Which instrument anchors price?** (See Remark 1.) Is the construction-cost shifter defensible against the existing sorting-model instruments, given the novelty bar?
- **Renters vs. owners.** The homevoter mechanism is about owners’ asset values; demand is over all households. Do we model tenure choice, or condition on it?

3 Layer II — The regulation game on two margins

This is the heart. The toy model’s homevoter chooses $(\bar{z}_j, \tau_j)$ to maximize $\Omega_j = \phi_j \lambda_j W_j^{\text{own}} + (1 - \phi_j \lambda_j) W_j^{\text{ent}}$. Several modeling choices have now been made; this section is rewritten around them. **The main specification is two coupled actions resolved by a within-jurisdiction two-stage (Stackelberg) game**, with the cross-jurisdiction spillover kept but the belief machinery deliberately stripped out. Two pieces of structure are worth separating cleanly at the outset, because they live at different levels and a reader (or referee) will conflate them if you do not:

- **Within a jurisdiction:** the two margins are set *sequentially* by two bodies — the council sets the slow map $\bar{z}_j$, then the planning commission sets the fast discretionary tax τ_j taking the realized map as given. This is a Stackelberg/backward-induction problem *internal* to j . No beliefs about anyone are required to solve it.
- **Across jurisdictions:** j ’s payoff depends on the submarket price $p_{s(j)}$, which aggregates co-submarket neighbors’ regulation. This is where the spillover — the empirical signature of the federalism breakdown — lives, and it is a *separate* modeling object from the within- j timing.

3.1 The within-jurisdiction two-stage game (main spec)

This is the committed structure and it is genuinely strong: it mirrors the institution exactly (two bodies, two speeds, sequential), and it makes the leakage object mechanical rather than assumed. Solve by backward induction:

$$\text{Stage 2 (commission): } \tau_j^*(\bar{z}_j) = \arg \max_{\tau_j} \Omega_j(\bar{z}_j, \tau_j; p_{s(j)}), \quad (2)$$

$$\text{Stage 1 (council): } \bar{z}_j^* = \arg \max_{\bar{z}_j} \Omega_j(\bar{z}_j, \tau_j^*(\bar{z}_j); p_{s(j)}). \quad (3)$$

The leakage primitive is then exactly the slope of the stage-2 best response, $\partial \tau_j^* / \partial \bar{z}_j$: when pre-emption forces $\bar{z}_j$ up (the map loosens), the commission re-optimizes τ_j upward in compensation. **This is why the two-stage structure is not cosmetic** — it is what makes Question 2 (leakage) a structural object readable off (2) rather than only a reduced-form DiD coefficient. The within- j game requires no information-structure assumption at all, which is what lets us honor the “no beliefs about others” simplification without losing the two-margin substance.

3.2 The cross-jurisdiction spillover: two readings to choose between

“Abstract away from beliefs about other localities” admits two genuinely different implementations, and the choice is yours to make. The memo lays out both; they are *not* notational variants — they imply different papers.

Reading A — spillover survives, via rational expectations over the equilibrium price.

Keep the cross-jurisdiction interaction, but replace *beliefs about individual neighbors’ actions* with *correct anticipation of the equilibrium submarket price*. Each jurisdiction is a price-taker in the regulation game: it takes $p_{s(j)}$ as given and rationally anticipated, where $p_{s(j)}$ is the fixed point that aggregates everyone’s regulation through the demand/clearing block of Layers I and III. The spillover operates through this *price aggregator*, not through bilateral best-response, so there is no belief-formation machinery, no CCP first step, and no incomplete-information estimation — yet the reaction function (and hence the empirical signature of the spillover) is preserved, now mediated by

the equilibrium price. This is the standard “competitive-in-the-externality, rational-expectations-equilibrium” move and it is the lightest way to keep the spillover alive. *If you want the reaction function to remain the centerpiece of the paper (as the toy model frames it), this is the reading.*

Reading B — drop cross-jurisdiction interaction entirely. Each jurisdiction solves its two-stage problem in isolation, taking its *own* local conditions as given and *not* responding to other jurisdictions at all. The spillover then enters the paper *only through the planner benchmark* in Layer III: the decentralized equilibrium is a collection of independent single-jurisdiction optimizations, and the Oates wedge is the gap between that and the internalizing planner’s allocation. This is maximally tractable and still answers Question 1 (the distortion) and Question 2 (leakage, off the within- j two-stage game). **What it gives up** is the reaction function as a *directly estimated* object — there is no strategic slope to estimate because jurisdictions do not, by assumption, react to each other. The toy model’s “nonzero strategic-interaction slope is the signature of the spillover” becomes a statement you *impose via the planner* rather than *measure*.

Remark 2 (The choice is really about what the empirical centerpiece is). *Reading A keeps “estimate the reaction function” as the headline empirical task and the direct test of the federalism breakdown; it costs you a fixed-point/equilibrium-price computation and a defensible price-taking assumption. Reading B makes the paper a pair of clean single-jurisdiction-plus-planner exercises; it is simpler and still delivers both headline numbers, but the spillover is assumed into the benchmark rather than read off a slope.* This is the single most consequential decision in Layer II and the first thing to settle with Guren.

3.3 What the spatial-lag reduced form still does (under either reading)

Independently of A vs. B, the spatial-lag regression remains the natural *first* empirical result and a descriptive diagnostic:

$$\tau_{jt} = \rho \sum_{k:s(k)=s(j)} w_{jk} \tau_{kt} + x_{jt}\beta + u_{jt}, \quad (4)$$

with $W = [w_{jk}]$ a co-submarket weight matrix, estimated by spatial 2SLS (Kelejian and Prucha, 1998). Under Reading A a nonzero ρ is the reduced-form shadow of the structural reaction function; under Reading B it is a specification check (you would want to understand any $\rho \neq 0$ you find, since your model assumes it away). Either way, ρ alone cannot separate the spillover from yardstick competition or common shocks (Brueckner, 1998, 2003) — which is exactly why the structural machinery (the planner benchmark, or the equilibrium-price reaction function) is needed for the counterfactual.

3.4 Action type: continuous vs. ordered vs. binary (open)

What *kind* of variable each action is shapes the estimator, and this is unresolved. The three options, even-handedly:

- **Continuous.** τ_j as built from the minutes is naturally a near-continuous tax index (process exposure, opposition counts, conditioning rates, delay), and $\bar{z}_j$ as an envelope measure is at least semi-continuous. *For:* preserves information, makes the two-stage FOCs (2)–(3) smooth and differentiable, gives a clean $\partial\tau_j^*/\partial\bar{z}_j$ leakage slope. *Against:* requires committing to a cardinalization of the tax index (what *is* one unit of τ ?), and continuous-action games have thinner off-the-shelf estimation theory than discrete ones.

- **Ordered.** Bin each margin into ordered categories (e.g. restriction tiers). *For:* robust to the exact cardinalization, maps to a sizeable ordered-game literature, keeps monotone structure. *Against:* discards within-tier variation; the leakage object becomes a discrete-response change rather than a smooth slope; bin cutoffs are themselves a judgment call.
- **Binary.** Reduce each margin to restrict/not (or above/below a threshold). *For:* maximal tractability, the entire discrete-games toolkit applies directly, cleanest identification arguments. *Against:* throws away most of what your hand-built minutes data *measures* — the whole point of the τ_j construction is its granularity — so a binary action wastes the project’s central data asset.

The tension is the usual information-vs-tractability one, sharpened here because your comparative advantage *is* the granular τ_j . A reasonable instinct (not a commitment): continuous τ_j to honor the data, with an ordered version as the robustness/identification-clean companion. Worth a deliberate decision with Guren rather than a default.

3.5 The preemption exclusion restriction (not yet committed)

Earlier drafts leaned on a specific exclusion restriction: that a *neighbor’s* preemption exposure shifts the neighbor’s action but does not enter j ’s payoff except through that action, which would identify a strategic slope. **This is now explicitly held open — do not commit to it without careful thought.** The reasons for caution are real:

- Preemption exposure is *spatially correlated* — Housing-Element non-compliance and builder’s-remedy exposure cluster, so a neighbor’s exposure and j ’s own exposure move together, which is exactly what an exclusion restriction needs them *not* to do.
- Exposure may enter j ’s payoff *directly* (a regional demand or political shock that hits the whole submarket), not only through the neighbor’s action.
- Under Reading B there is no strategic slope to identify, so this exclusion restriction is moot; under Reading A it is load-bearing. So the exclusion question is downstream of the A/B choice — another reason to settle that first.

What the preemption ramp can *more safely* do, regardless, is power the within- j leakage DiD (Question 2): comparing a jurisdiction’s own $\bar{z}_j$ and τ_j before/after its own exposure, which needs only that exposure timing is as-good-as-random with respect to the jurisdiction’s *contemporaneous* demand shock — a weaker and more defensible claim than the cross-neighbor exclusion. Treat the cross-jurisdiction exclusion restriction as a research question, not an assumption.

3.6 Multiplicity and equilibrium selection (open)

Also unresolved, and the right answer depends heavily on A vs. B:

- **Under Reading B**, the within- j two-stage game is a single-agent sequential optimization; with a concave-enough objective it has a unique solution and *multiplicity largely does not arise*. This is a substantial simplicity dividend of Reading B and worth weighing in the A/B decision.
- **Under Reading A**, the equilibrium is a fixed point in the submarket price; uniqueness depends on the curvature of demand and the disamenity, and multiple price equilibria are possible in principle. If they arise, the standard responses are (i) impose a selection rule (e.g. the equilibrium nearest the observed data, as in the quantitative-spatial inversion of Layer III), or (ii) set-identify and report bounds (Bajari et al., 2010b, de Paula, 2013).
- A practical first pass under either reading: check uniqueness numerically at the estimated parameters (solve from many starting points), and only invest in selection/bounds machinery

if multiplicity actually shows up.

For Guren — Layer II, updated

- **Reading A vs. B (Remark 2)** — the decision that defines the paper. Keep the reaction function as an estimated centerpiece (A, via rational expectations over the equilibrium price), or make the paper single-jurisdiction-plus-planner and let the spillover enter only through the benchmark (B)? Everything else in Layer II is downstream of this.
- **Action type** (§3.4): continuous τ_j honors the data but needs a cardinalization; ordered/binary are more tractable but waste granularity. No commitment yet.
- **The cross-neighbor exclusion restriction** (§3.5) is *not* assumed. Is it salvageable given spatial clustering of preemption exposure, or do we lean on the within- j leakage DiD instead (which needs only own-exposure-as-good-as-random)?
- **Multiplicity** (§3.6): under B largely a non-issue; under A, check uniqueness numerically first and only then decide between a selection rule and set identification.
- **Does Question 2 even need the cross-jurisdiction game?** Leakage is identified off the within- j two-stage structure plus the own-exposure DiD, so it may stand under *either* reading. Worth confirming that only Question 1’s reaction-function framing forces the A/B choice.

4 Layer III — Closing the spatial equilibrium

Demand (Layer I) and the regulation game (Layer II) each take the *other side* as given. To compute counterfactuals — regulation chosen by an internalizing planner; a preemption shock that propagates — you need a general equilibrium that closes the loop: regulation $\rightarrow$ supply $\rightarrow$ prices and density $\rightarrow$ wages via agglomeration $\rightarrow$ population $\rightarrow$ back into demand and into the homevoter’s asset value.

4.1 The minimal closure

Adopt the quantitative-spatial template (Ahlfeldt et al., 2015, Redding and Rossi-Hansberg, 2017): locations with labor-market access, floor-space supply, and idiosyncratic location tastes (the Layer I logit *is* the residential-choice block of such a model — they are the same discrete-choice kernel, which is the clean conceptual bridge between the IO and spatial literatures). Concretely you need:

1. a **housing supply function** mapping the regulatory pair $(\bar{z}_j, \tau_j)$ to dwellings H_j — this is where r_j bites on quantities;
2. an **agglomeration/wage block** $w_m = w(N)$ (can be calibrated, not estimated, to a literature elasticity — a place to *borrow*);
3. **market clearing** delivering p_s and D_j consistent with the toy model’s $p_s = \alpha N_s / H_s$.

4.2 Inversion and counterfactuals

Quantitative-spatial models are estimated by *inversion*: assume observed data is an equilibrium, back out unobserved location productivities/amenities (A_j, B_j) that rationalize it, then re-solve under the counterfactual (often via exact-hat algebra to avoid re-estimating levels). The Oates wedge (Question 1) is then: invert to recover primitives $\rightarrow$ replace the decentralized Nash regulation with the internalizing planner’s choice $\rightarrow$ re-solve $\rightarrow$ compare aggregate H, p , welfare. Preemption

leakage (Question 2) is: apply the preemption shock to $\bar{z}_j$, let the Layer II game re-equilibrate τ_j , re-solve, and read off the fraction of intended loosening clawed back.

For Guren — Layer III

- **How much GE do you actually need?** Full ARSW-style inversion is a large undertaking. A *partial-equilibrium* submarket-clearing closure (prices respond to quantities, wages held fixed) may suffice for both headline questions, with full spatial GE as an extension. Guren will have a strong, fast view on the minimum viable closure — this is the highest-leverage scoping question in the whole note.
- **Calibrate or estimate agglomeration?** Almost certainly calibrate $w(N)$ to an existing elasticity rather than identify it yourself; confirm.
- **Exact-hat algebra vs. levels.** Can the counterfactuals be done in changes (cheap, robust) rather than recovering all levels?
- **Where does the supply function $H_j(\bar{z}_j, \tau_j)$ come from?** This is the join between the regulation game and the spatial block, and it is currently the least-specified map in the whole architecture.

5 How the three layers compose

Layer	Recovers	Method / literature
I. Demand	WTP distribution; $\partial p_s / \partial H_s$; price elasticities	RC logit / sorting; BLP contraction; cost-shifter IV (Berry et al., 1995 , Bayer et al., 2007 , Conlon and Gortmaker, 2020)
II. Regulation game	homevoter weights $\phi_j \lambda_j$; disamenity γ ; margin-substitution $\partial \tau_j^* / \partial \bar{z}_j$ (leakage); reaction function (Reading A only)	within- j two-stage Stackelberg; rational-expectations equilibrium price (A) or single-juris. + planner (B); own-exposure DiD for leakage (Bajari et al., 2010a , Seim, 2006)
III. Spatial closure	location primitives (A_j, B_j) ; equilibrium p, D, N, w	QSE inversion; calibrated agglomeration; exact-hat counterfactuals (Ahlfeldt et al., 2015 , Redding and Rossi-Hansberg, 2017)

The estimation order is bottom-up then close: estimate demand (I); estimate the game (II) taking demand-implied price responses as inputs; assemble the supply function and invert the spatial block (III); compute counterfactuals by perturbing II's equilibrium and re-solving III. The two headline questions are then read off the re-solved equilibrium.

6 The one slide that should drive the meeting

If the meeting does one thing, it should pin down the **minimum viable version of each layer**, because the ambitious version of all three at once is a multi-year program, not a three-week exploration. A defensible “minimum viable paper” is:

- **Demand:** RC sorting with a construction-cost instrument (your IO contribution), at the jurisdiction or coarse-neighborhood level.
- **Regulation:** the within- j two-stage Stackelberg game (council $\rightarrow$ commission) as the committed spec, with the two-margin leakage identified via the own-exposure DiD around preemption; the spatial-lag reduced form (4) as the descriptive first result. The one genuinely open fork is Reading A (keep the reaction function, via rational expectations over the equilibrium price) vs. Reading B (single-jurisdiction + planner benchmark).
- **Closure:** partial-equilibrium submarket clearing with calibrated agglomeration; full QSE inversion flagged as the extension.

Everything above this line is the paper; everything below (full QSE, dynamic ratchet, and — if Reading A’s equilibrium-price computation proves too costly — a retreat to Reading B) is the extension/fallback frontier, consistent with the toy model’s own §9 scoping.

Ranked questions to walk in with.

1. **Reading A vs. B** (Remark 2) — does the reaction function stay an estimated centerpiece, or does the spillover enter only through the planner benchmark? Defines the paper.
2. Minimum viable *closure* (Layer III) — biggest scope lever, and entangled with A/B (Reading A needs the equilibrium-price fixed point).
3. Whether the cross-neighbor *exclusion restriction* on preemption is salvageable, or leakage leans on the within- j own-exposure DiD instead (§3.5).
4. *Action type* — continuous vs. ordered vs. binary (§3.4).
5. The *supply function* $H_j(\bar{z}_j, \tau_j)$ that joins II and III.

References

- Gabriel M. Ahlfeldt, Stephen J. Redding, Daniel M. Sturm, and Nikolaus Wolf. The economics of density: Evidence from the berlin wall. *Econometrica*, 83(6):2127–2189, 2015.
- Patrick Bajari, Han Hong, John Krainer, and Denis Nekipelov. Estimating static models of strategic interactions. *Journal of Business & Economic Statistics*, 28(4):469–482, 2010a.
- Patrick Bajari, Han Hong, and Stephen P. Ryan. Identification and estimation of a discrete game of complete information. *Econometrica*, 78(5):1529–1568, 2010b.
- Patrick Bayer, Robert McMillan, and Kim S. Rueben. An equilibrium model of sorting in an urban housing market. *NBER Working Paper No. 10865*, 2004.
- Patrick Bayer, Fernando Ferreira, and Robert McMillan. A unified framework for measuring preferences for schools and neighborhoods. *Journal of Political Economy*, 115(4):588–638, 2007.
- Steven Berry, James Levinsohn, and Ariel Pakes. Automobile prices in market equilibrium. *Econometrica*, 63(4):841–890, 1995.
- Steven T. Berry. Estimating discrete-choice models of product differentiation. *RAND Journal of Economics*, 25(2):242–262, 1994.
- Jan K. Brueckner. Testing for strategic interaction among local governments: The case of growth controls. *Journal of Urban Economics*, 44(3):438–467, 1998.
- Jan K. Brueckner. Strategic interaction among governments: An overview of empirical studies. *International Regional Science Review*, 26(2):175–188, 2003.

- Christopher Conlon and Jeff Gortmaker. Best practices for differentiated products demand estimation with pyblp. *RAND Journal of Economics*, 51(4):1108–1161, 2020.
- Áureo de Paula. Econometric analysis of games with multiple equilibria. *Annual Review of Economics*, 5:107–131, 2013.
- Amit Gandhi and Jean-François Houde. Measuring substitution patterns in differentiated-products industries. *Working Paper / Econometrica (rev.)*, 2019. Differentiation IVs.
- Harry H. Kelejian and Ingmar R. Prucha. A generalized spatial two-stage least squares procedure for estimating a spatial autoregressive model with autoregressive disturbances. *Journal of Real Estate Finance and Economics*, 17(1):99–121, 1998.
- Aviv Nevo. A practitioner’s guide to estimation of random-coefficients logit models of demand. *Journal of Economics & Management Strategy*, 9(4):513–548, 2000.
- Stephen J. Redding and Esteban Rossi-Hansberg. Quantitative spatial economics. *Annual Review of Economics*, 9:21–58, 2017.
- Albert Saiz. The geographic determinants of housing supply. *Quarterly Journal of Economics*, 125(3):1253–1296, 2010.
- Katja Seim. An empirical model of firm entry with endogenous product-type choices. *RAND Journal of Economics*, 37(3):619–640, 2006.